

CMT83086 High Reliability Isolated Full Duplex RS-485 Transceiver

1 Characteristic

- Safety-related certifications
 - VDE certification: DIN VDE V 0884-11: 2017-01
 - UL1577 certification: 5000Vrms
 - CSA Certification: Components Comply with 5A
 - CQC Certification: Compliant with GB4943.1-2011
- Insulation withstand voltage is up to 5000Vrms
- Bus side power supply voltage: 3.3 V to 5.5 V
- VDD1 power supply voltage: 2.5 to 5.5 V
- High CMTI: ± 200 kV/us
- Higher system-level EMC performance:
- Bus Pin compliant with IEC61000-4-2 ± 12 kV ESD
 - Other pins comply to ± 7 kV contact ESD
 - Operating temperature: -40 °C to 125 °C
- Fail-safe receiver
- Isolation barrier life:
 - Over 40 years
- Supports 256 transceivers
- Compliant with RoHS package SOIC 16 (wide body)

2 Applications

- Work automation system
- Isolation 485 Communication System
- Smart electricity and water meters
- Security and monitoring systems

3 Description

The CMT83086 is a highly reliable full-duplex RS-485 transceiver based on CMOSTEK digital isolation technology. CMT83086 has passed UL1577 safety certification, supports 5kVrms insulation withstand voltage, and has low radiation, low power consumption and strong anti-electromagnetic interference capabilities.

The bus pin on the bus side of the CMT83086 has ± 12 kV System-level ESD protection to the ground. It has the is fail-safe function. When the receiver input is open or shorted, the receiver output logic is ensured to be high. It has a receiver input impedance with 1/8 unit load, allowing up to 256 transceivers to be connected on the bus.

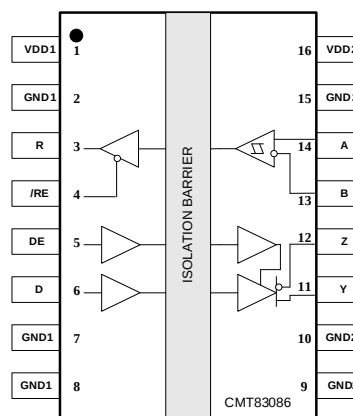
The data rate of the CMT83086 is 12Mbps. The device can reduce the electromagnetic interference of the transmission line at the wrong end.

The CMT83086 adopts wide-body SOIC-16 package.

Chip Information

Part Number	Package	Dimension (mm x mm)
CMT83086	WB (W) SOIC-16	10.4 x 7.5
Refer to Section 12 for more ordering information.		

Functional Block Diagram



Catalogue

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4 Pin Configuration and Function Description

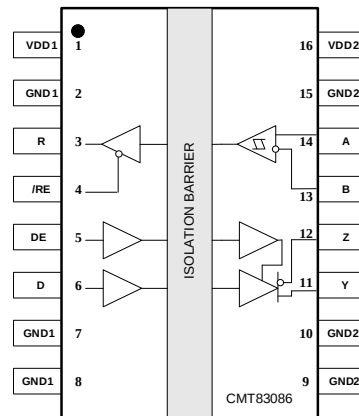


Figure 4-1. CMT83086 Pin Diagram

Pin number	Pin Name	Functional Description
1	VDD1	Isolation Side 1 Supply Voltage
2	GND1	Ground reference point of isolation side 1
3	R	Receive output
4	/RE	Receive input enabled. This port is low level active
5	DE	Drive input enabled. This port is high level active
6	D	Driver transmitting data input
7	GND1	Ground reference point of isolation side 1
8	GND1	Ground reference point of isolation side 1
9	GND2	Ground reference point of Isolation side 2
10	GND2	Ground reference point of isolation side 2
11	Y	Driver non-inverting output. When the driver is disabled, or VDD1/ VDD2 is powered down, pin Y goes into a high impedance state to avoid bus overloading.
12	Z	Driver inverting output. When the driver is disabled, or when VDD1/ VDD2 is powered down, pin Z goes into a high impedance state to avoid bus overloading.
13	B	Receiver inverting input
14	A	Receiver non-inverting input
15	GND2	Ground reference of isolation side 2
16	VDD2	Supply voltage to isolation side 2

5 Specifications

5.1 Absolute Maximum Rating

The parameters are tested within the working temperature range under the condition of natural ventilation (unless otherwise noted)⁽¹⁾

Table 5-1. Absolute maximum rating

Parameter	Symbol	Min	Max	Unit
Power supply voltage ^[2]	VDD ₁ , VDD ₂	-0.3	6	V
Voltage of bus I/O pin: A, B, Y, Z	V _O	-8	13	V
Voltage input, Transient pulse, pin A, B, Y and Z (Through 100 Ω resistance, see Figure 9-11)	V _{IT}	-50	50	V
Voltage input on any D, DE or RE port	V _I	-0.5	6	V
Receiver output current	I _O	-10	10	mA
Junction temperature	T _J		150	°C
Storage temperature	T _{STG}	-65	150	°C

(1) Stresses exceeding the values listed under the absolute maximum rating may cause permanent damage to the device. These listed values are stress ratings only, which do not indicate that the device will operate properly under these conditions and under any other conditions under the recommended operating conditions. Extended period time under absolute maximum rating conditions may affect the reliability of the device.

5.2 ESD Rating

Parameter			Value	Unit
V _(ESD) Electrostatic discharge	Human Body Discharge Model (HBM), apply for ANSI/ESDA/JEDEC JS-001 standard ⁽¹⁾	Bus pin and GND1	±6000	V
		Bus pin and GND2	±12000	V
		All pins	±4000	V
	Charger device model (CDM), JEDEC specification JESD22-C101, all pins ⁽²⁾		±1000	V

(1) JEDEC document JEP155 states that 500V HBM can achieve safe production under standard ESD control process.
(2) JEDEC document JEP157 states that 250V CDM can achieve safe production under standard ESD control process.

5.3 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
V _{DD1}	Logic side supply voltage	2.7		5.5	V
V _{DD2}	Bus side supply voltage	2.9	5	5.5	V
V _{OC}	Voltage on bus I/O pin A/B	-7		12	V
V _{IH}	High input voltage (D, DE, RE input)	2		V _{DD}	V
V _{IL}	Low level input voltage (D, DE,	0		0.8	V

	RE input)				
V_{ID}	Differential input voltage	-12		12	V
R_L	Differential load resistance	54	60		Ω
I_O	Output current, driver	-60		60	mA
I_O	Receiver output current	-10		10	mA
T_A	Operating ambient temperature	-40		125	$^{\circ}\text{C}$

5.4 Thermal Performance Information

Symbol	Thermal index ⁽¹⁾	Unit
$R_{\theta JA}$	Junction to ambient thermal resistance	79.6 $^{\circ}\text{C/W}$
$R_{\theta JC (top)}$	Junction to chip housing (top) thermal resistance	39.7 $^{\circ}\text{C/W}$
$R_{\theta JB}$	Junction to circuit board thermal resistance	44.7 $^{\circ}\text{C/W}$
ψ_{JT}	Junction to top characteristic parameter	13.2 $^{\circ}\text{C/W}$
ψ_{JB}	Junction to circuit board characteristic parameter	44.0 $^{\circ}\text{C/W}$
$R_{\theta JC (bot)}$	Junction to chip housing (bottom) thermal resistance	— $^{\circ}\text{C/W}$

5.5 Insulation Specifications

Parameter	Test conditions	Value	Unit
CLR	External clearance ⁽¹⁾	Shortest spatial distance between terminals	8 mm
CPG	External creepage distance ⁽¹⁾	Shortest package surface distance between terminals	8 mm
DTI	Insulation penetration distance	Minimum internal clearance	8 μm
CTI	Relative leakage tracing index	DIN EN 60112 (VDE 0303-11); IEC 60112	>400 V
	Material Group	Conform to IEC 60664-1	II
	Over-voltage category (per IEC 60664-1)	Rated mains voltage $\leq 150 V_{RMS}$	I-IV
		Rated mains voltage $\leq 300 V_{RMS}$	I-III
DIN ENIEC 60747-17 (VDE 0884-17)⁽²⁾			
V_{IORM}	Maximum repeated peak isolation voltage	AC voltage (bipolar)	1414 V_{PK}
V_{IOTM}	Maximum transient isolation voltage	$V_{TEST} = V_{IOTM}$, $t = 60\text{s}$ (Identification test); $V_{TEST} = 1.2 \times V_{IOTM}$, $t = 1\text{s}$ (100% production test)	7000 V_{PK}
q_{pd}	Apparent charge ⁽³⁾	Method b; Routine test (100% production test) $V_{ini} = 1.2 \times V_{IOTM}$, $t_{ini} = 1\text{s}$; V_{pd} $(m) = 1.5 \times V_{IORM}$, $t_m = 1\text{s}$	≤ 5 pC
C_{IO}	Barrier capacitance, input to output ⁽⁴⁾	$V_{IO} = 0.4 \times \sin(2\pi ft)$, $f = 1\text{ MHz}$	2 pF
C_I	Ground input capacitance	$V_I = V_{DD}/2 + 0.4 \times \sin(2\pi ft)$, $f = 1\text{ MHz}$, $V_{DD} = 5\text{V}$	2 pF
R_{IO}	Isolation resistance ⁽⁴⁾	$V_{IO} = 500\text{V}$, $T_A = 25\text{ }^{\circ}\text{C}$	>1012 Ω
		$V_{IO} = 500\text{V}$, $T_S = 150\text{ }^{\circ}\text{C}$	>109 Ω
	Contamination level		2
	Climate category		40/125/21

UL 1577				
V_{ISO}	Maximum withstand isolation voltage	$V_{TEST} = V_{ISO}$, $t = 60s$ (Identification test), $V_{TEST} = 1.2 \times V_{ISO}$, $t = 1s$ (100% production test)	5000	V_{RMS}
(1) The creepage distance and clearance should meet the application requirements of the specific equipment isolation standards. Please be careful to maintain the creepage distance and clearance of the circuit board design to ensure that the mounting pads of the isolator on the printed circuit board do not shorten the distance. In particular cases, the creepage distance are equal to the gap on the printed circuit board. Employing techniques such as insert grooves or ribs on printed circuit boards can help improve the mentioned specifications. (2) This coupler is only suitable for basic electrical insulation within the safety rating range. The safety ratings shall be guaranteed by suitable protection circuits. (3) Apparent charge is Electrical discharge caused by partial discharge(pd). (4) Connect all the pins on each side of the isolation barrier to form a two-terminal device.				

5.6 Safety Limited Value

Safety restrictions⁽¹⁾ are aimed at minimizing potential damage to the isolation barrier in the event of input or output circuit failure.

Parameter	Test conditions	Min	Typ	Max	Unit
I_S	Safety input, output, or supply current			286	mA
T_S	Maximum safe temperature			150	°C

- 1) The maximum safe operating temperature T_S is the same as the device specified maximum junction temperature T_J . The I_S and P_S parameter represent the safe current and safe power respectively. Do not exceed the maximum limit of I_S and P_S . These limits vary with ambient temperature T_A .
- 2) The device of junction to air thermal resistance $R_{\theta JA}$ is mounted on the corresponding high K test board of the lead type surface package. Parameters can be calculated by the following formula:
- 3) $T_J = T_A + R_{\theta JA} \times P$, among which, P is the device power consumption.
- 4) $T_{J(max)} = T_S = T_A + R_{\theta JA} \times P_S$, where $T_{J(max)}$ is the maximum junction temperature.
- 5) $P_S = I_S \times V_I$, Among which, V_I is the maximum input voltage.

6 Electrical Characteristics

6.1 Driver Electrical Characteristics

Unless otherwise stated, all parameters are measured under the following conditions, $V_{DD1} = 3.3V$, $V_{DD2} = 5V$, $/RE = V_{DD1}$, $CL = 0$, and the load capacitance of the external oscilloscope probe of the test item with waveform output is about 15pF.

Table 6-1. Electrical Characteristics

Symbol	Description	Test condition	Min	Typ	Max	Unit
$ V_{OD} $	Driver Differential Output Voltage Amplitude	$D = DE = V_{DD}$, $I_O = 0mA$, no load	4.0	4.9	5.9	V
		$D = GND$, $DE = V_{DD}$, $I_O = 0mA$, no load	4.0	5.0	5.9	V
		$D = DE = V_{DD}$, $R_L = 54\Omega$, see Figure 9-1	2.5	3.1	3.7	V

		D = DE = VDD, $R_L = 100\ \Omega$, see Figure 9-1	2.9	3.6	4.3	V
$\Delta V_{OD} $	Differential output voltage variation between two states	D = GND or D = VDD, differential output voltage between two states, DE = VDD, no load,	3.2	4.0	4.8	mV
V_{OC}	Common mode output voltage VOC1	D = GND, DE = VDD, $R_L = 2 * R = 54\ \Omega$	2.0	2.5	3.0	V
	Common mode output voltage VOC2	D = VDD, DE = VDD, $R_L = 2 * R = 54\ \Omega$	2.0	2.5	3.0	V
$\Delta V_{OC(SS)}$	Steady-state common-mode output voltage variation between two states	$\Delta V_{OC(SS)} = VOC1 - VOC2$	-8.0	-10.0	-12.0	mV
$V_{OC(PP)}$	Peak-to-peak common mode output voltage (with oscilloscope probe)	DE = VDD, D = 100Khz square wave, $V_{PP} = 3V$, $Y_L = Z_L = 27\ \Omega$, see Figure 9-3	0.3	0.4	0.4	V
I_I	Logic input current	DE = D = 0, measure the current of D, and connect the ammeter to D and GND	-1.9	-2.3	-2.8	μA
I_{OZ}	High impedance state output current	$V_Z = 12V$, VDD = 0V, DE = D = GND, measure Z pin current	10.9	13.6	16.3	μA
		$V_Z = 12V$, VDD = 5V, DE = D = GND, measure Z pin current	10.5	13.1	15.8	μA
		$V_Y = 12V$, VDD = 0V, DE = D = GND, measure the Z pin current	10.9	13.6	16.3	μA
		$V_Y = 12V$, VDD = 5V, DE = D = GND, measure the Z pin current	10.5	13.1	15.7	μA
		$V_Y = -7V$, VDD = 0V, DE = D = GND, measure the Z pin current	-2.0	-2.5	-3.0	μA
		$V_Y = -7V$, VDD = 5V, DE = D = GND, measure the Z pin current	-2.2	-2.7	-3.3	μA
		$V_Z = -7V$, VDD = 0V, DE = D = GND, measure Z pin current	-1.6	-2.0	-2.4	μA
		$V_Z = -7V$, VDD = 5V, DE = D = GND, measure Z pin current	-3.7	-4.6	-5.5	μA
I_{OS}	Short circuit output current	Z=Y=12V, DE=VDD, D=VDD, $V_Y = HIGH$	49.7	62.2	74.6	mA
		Z = Y = -7V, DE = VDD, D = GND, $V_Y = LOW$	-59.6	-74.5	-89.4	mA

6.2 Receiver Electrical Characteristics

Unless otherwise specified, the parameters are measured under VDD1 = 3.3 V and VDD2 = 5V, where DE = GND, CL = 15pF, and the load capacitance of the test item with waveform output external oscilloscope probe is approximately 15pF.

Table 6-2. Receiver Electrical Characteristics

Parameter	Description	Test condition	Min	Typ	Max	Unit
I_{IN2}	Bus input current (A, B)	$V_{DD} = 5V, V_{IN} = 12V$ V_A or $V_B = 12V$	0.05	0.06	0.07	mA
		V_A or $V_B = 12V, V_{DD} = 0$, other inputs are 0	0.05	0.06	0.07	mA
		V_A or $V_B = -7V, V_{DD} = 5V$, other inputs are 0	0.03	0.04	0.05	mA
		V_A or $V_B = -7V, V_{DD} = 0$, other inputs are 0	0.03	0.04	0.04	mA
V_{IT+}	Forward input threshold voltage	$-7V \leq V_{CM} = 0 \leq 12V, A = GND$	-58.72	-73.40	-88.08	mV
V_{IT-}	Negative input threshold voltage	$-7V \leq V_{CM} = 0 \leq 12V$	-85.12	-106.40	-127.68	mV
V_{hys}	Input hysteresis	$V_{hys} = (V_{IT+} - V_{IT-})$	26.40	33.00	39.60	mV
V_{IT+}	Forward input threshold voltage	$V_{CM} = 12V, A = 12V,$ $B = 11.5-12.5V$	-32.80	-41.00	-49.20	mV
V_{IT-}	Negative input threshold voltage		-48.80	-61.00	-73.20	mV
V_{hys}	Input hysteresis		16.00	20.00	24.00	mV
V_{IT+}	Forward input threshold voltage	$V_{CM} = 12V, A = -7V,$ $B = -7.5 \sim -6.5V$	-74.40	-93.00	-111.60	mV
V_{IT-}	Negative input threshold voltage	$V_{hys} = (V_{IT+} - V_{IT-})$	-91.20	-114.00	-136.80	mV
V_{hys}	Input hysteresis		16.80	21.00	25.20	mV
V_{hys}	Input hysteresis	$V_{hys} = (V_{IT+} - V_{IT-})$	26.40	33.00	39.60	mV
V_{OH}	High-level output voltage	$V_{ID} = 200mV, I_O = -4mA$,	2.45	3.06	3.67	V
V_{OL}	Low level output voltage	$V_{ID} = -200mV, I_O = 4mA$	0.15	0.19	0.23	V
$I_{O(Z)}$	Output high impedance current on R pin (tri-state input current)	$0.4V < V_O < 2.4V, RE = High,$ $VR = 2V$, measure the current through R pin	0.08	0.10	0.12	uA

R_{IN}	Receiver input resistance	$V_B = 0V, V_A = 2V, -7 < V_{CM} < 12V$	217.28	271.60	325.92	k Ω
I_{OSR}	Receiver short circuit current	$V_{AB} = 1V, 0V < V_O < V_{DD}, RE = 0$, the R to ground current	42.18	52.72	63.26	mA

6.3 Supply Current

Unless otherwise specified, the parameters are measured at $V_{DD1} = 3.3V$ and $V_{DD2} = 5V$.

Parameter	Symbol	Test condition	Min	Typ	Max	Unit
Supply current	ICC1	Logic side supply current, $/RE = 0, DE = 0V$	1.5	1.9	2.3	mA
	ICC2	Bus side supply current, $/RE = 0, DE = 0V$	1.7	2.2	2.6	mA
	ICC1	Logic side supply current, $/RE = V_{DD}, DE = V_{DD}$	2.3	2.9	3.5	mA
	ICC2	Bus side supply current, $/RE = V_{DD}, DE = V_{DD}$	1.7	2.2	2.6	mA
	ICC1	Logic side supply current, $/RE = 0, DE = V_{DD}$	2.3	2.9	3.5	mA
	ICC2	Bus side supply current, $/RE = 0, DE = V_{DD}$	1.7	2.2	2.6	mA
Switch-off current	ISHDN1	Logic side current, $/RE = V_{DD}, DE = 0V$	1.5	1.9	2.3	mA
	ISHDN2	Bus side current, $/RE = V_{DD}, DE = 0V$	1.7	2.2	2.6	mA

6.4 Driver Switching Characteristics

Unless otherwise specified, the following parameters are measured under the conditions of $V_{DD1} = 3.3V$ and $V_{DD2} = 5V$, where the load capacitance $C_L = 0$, and the oscilloscope capacitance is approximately 15pF measured by oscilloscope probe.

Parameter	Symbol	Test condition Refer to Figure 9-2/9-3/9-4/9-5 for test method	Min	Typ	Max	Unit
Propagation Delay (driver input low to high)	t_{PLH}	$R_{DIFF} = 54\Omega$, D is connected to square wave, $F = 100KHz$, measure 50% of the input and output rising edge	0.8	0.9	1.1	ns
Propagation delay (Driver input from high to low)	t_{PHL}	$R_{DIFF} = 54\Omega$, D is connected to square wave, $F = 100KHz$, measure 50% of the input and output falling edge	0.6	0.7	0.9	ns

Pulse width distortion	PWD	$ t_{PHL}-t_{PLH} $	1.0	1.2	1.5	ns
Differential output rising edge time	t_r	$R_{DIFF}= 54\Omega$, D is connected to square wave, Freq = 100KHz, VPP = 3V (oscilloscope probe the input and output)	1.8	2.2	2.6	ns
Differential output falling edge time	t_f	$R_{DIFF}= 54\Omega$, D is connected to square wave, F = 100KHz, VPP = 3V (oscilloscope probe the input and output)	1.0	1.2	1.5	ns
Differential output rising edge time	t_r	$R_{DIFF}= 54\Omega$, D is connected to square wave, Freq = 100KHz, VPP = 3V (coaxial line input and output)	0.2	0.3	0.4	ns
Differential output falling edge time	t_f	$R_{DIFF}= 54\Omega$, D is connected to square wave, F = 100KHz, VPP = 3V (coaxial line input and output)	0.2	0.2	0.2	ns
Propagation delay, high impedance to high level output (enable to output high)	t_{PZH}	DE is connected to square wave, D = VDD, Y is connected to 110 Ω pull-down, Y output is in phase with input, and are measured by the delay of input rising edge 50% and output rising edge 50%	0.8	1.0	1.2	ns
Propagation delay, high level to high impedance output (input high to disabled)	t_{PHZ}	DE is connected to square wave, D = VDD, Y is connected to 110 Ω pull-down, Y output is in phase with input, measured by 50% of the input falling edge and 90% of the output falling edge delay	0.8	1.0	1.2	ns
Propagation delay, high impedance to low level output (enable to output low)	t_{PZL}	DE is connected to square wave, D = VDD, Z is connected to 110 Ω pull-up, Z output is inverted with input, measured by the delay between 50% of the input rising edge and 50% of the output falling edge	1.4	1.7	2.0	ns
Propagation delay, low level to high impedance output (input low to disabled)	t_{PLZ}	DE is connected to square wave, D = VDD, Z is connected to 110 Ω pull-up, Z output is inverted with input, measured by delay between 50% of the input falling edge and 10% of the output rising edge	1.9	2.4	2.9	ns

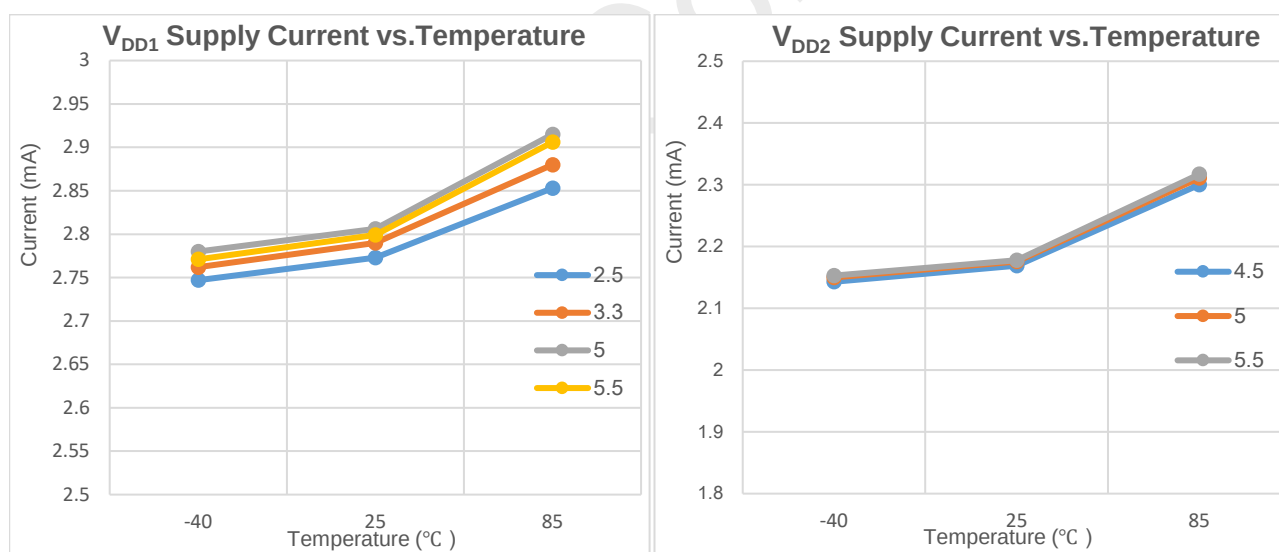
6.5 Receiver Switching Characteristics

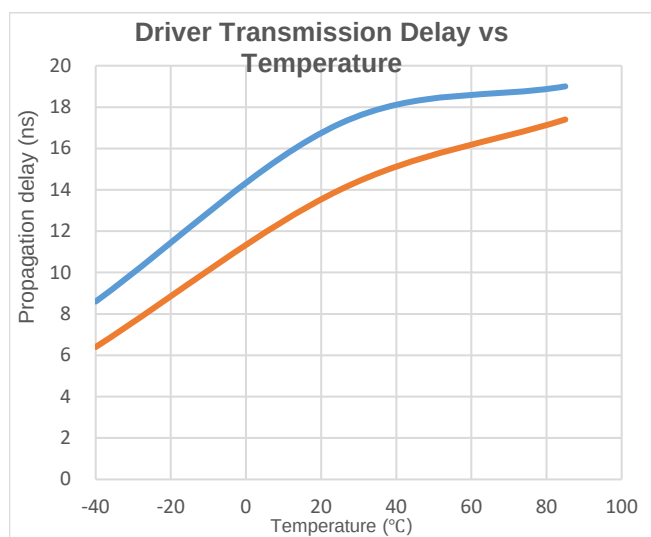
Unless otherwise specified, the following parameters are measured under the conditions of VDD1 = 3.3 V and VDD2 = 5V, where DE = D = suspended, and measured by the oscilloscope probe, the oscilloscope capacitance is approximately 15pF, that is, CL = 15pF.

Parameter	Symbol	Test condition Refer to Figure 9-7/9-8/9-9 for test method	Min	Typ	Max	Unit
Propagation Delay (driver input low to high)	t_{PLH}	B = 1.5 V, A is connected to square wave, frequency F = 100KHz, VPP = 3V, CL = 15pF (Both input and output are connected to oscilloscope probe)	66.4	83.0	99.6	ns
Propagation delay (Driver input from high to low)	t_{PHL}		68.8	86.0	103.2	ns
Pulse width distortion	PWD		2.4	3.0	3.6	ns
Differential output rising edge time	t_r		28.4	35.6	42.7	ns

Differential output falling edge time	t_f		33.0	41.2	49.4	ns
Differential output rising edge time	t_r	Both input and output are connected to coaxial axis	2.1	2.6	3.1	ns
Differential output falling edge time	t_f	Both input and output are connected to coaxial axis	2.0	2.5	3.0	ns
Propagation delay, high level to high impedance output	t_{PHZ}	A = 1.5 V, B = 0V, RE/= 100KHz, vpp = 3V, RO connects 1K Ω resistor to the ground, CL = 15pF, test point VI rising edge = 50%, VO falling edge = 90%	3.2	4.0	4.8	ns
Propagation delay, high impedance to high output	t_{PZH}	A = 1.5 V, B = 0V, RE/= 100Khz, vpp = 3V, RO connects 1K Ω resistor to ground, CL = 15PF, test point VI falling edge = 50%. VO rising edge = 50%	31.0	38.8	46.6	ns
Propagation delay, low to high impedance output	t_{PLZ}	A = 0V, B = 1.5 V, RE/= 100Khz, vpp = 3V, RO connects 1K Ω resistor to VDD, CL = 15pF, test point VI rises 50%, VO rises 10%	3.2	4.0	4.8	ns
Propagation delay, high impedance to low level output	t_{PZL}	A = 0V, B = 1.5 V, RE/= 100Khz, vpp = 3V, RO connects 1K Ω resistor to VDD, CL = 15pF, and test point VI falling edge is 50%. VO falling edge 50%	29.6	37.0	44.4	ns

6.6 Typical Performance





7 Safety Certification

Table 7-1. Safety certification

VDE	UL	CSA	CQC
DIN VDE V0884-11: 2017-01	UL 1577 Component Recognition Program	Approved under CSA Component Acceptance Notice 5A	GB 4943.1-2011
Certificate number: pending	Certificate number: UL-US-2439077-1	Certificate number: UL-CA-2429797-1	Certificate number: pending

8 Typical Applications

8.1 Typical Application Schematic

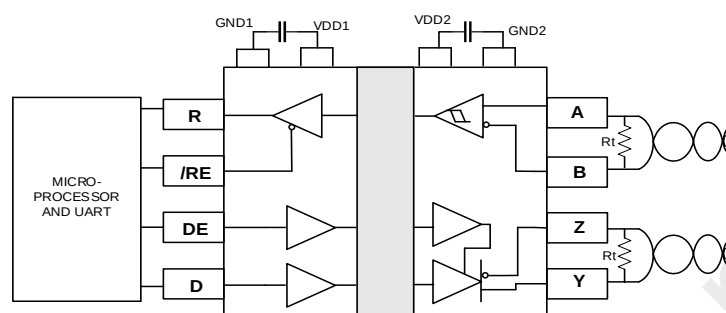
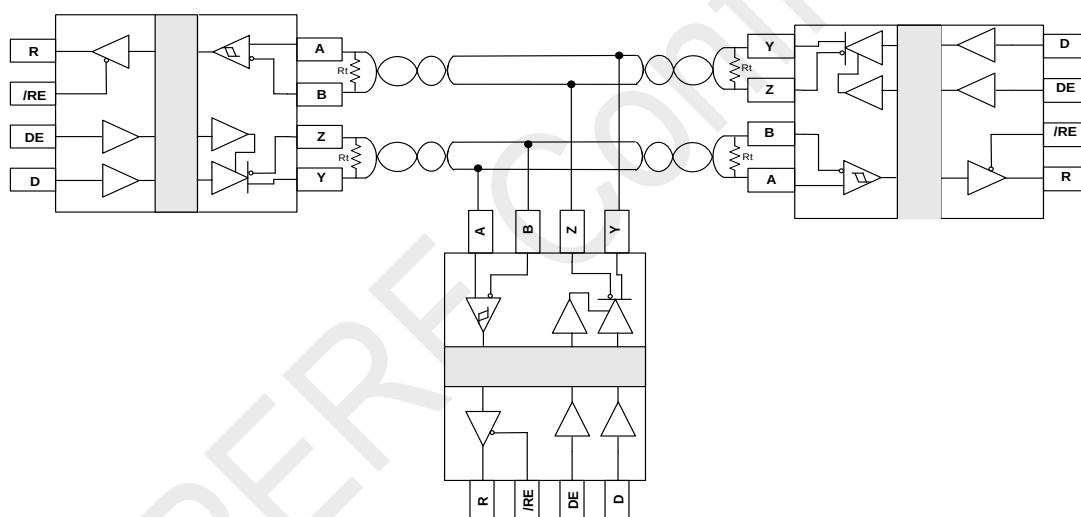


Figure 8-1. Typical application schematic



Note: R_t is a special matched impedance, typically it is 120 Ω .

Figure 8-2. Typical Isolated Full-duplex RS-485 Application

8.2 PCB Layout

CMT83086 requires a 0.1 μ F bypass capacitor from VDD1 to GND1, while VDD2 to GND2 requires a 10 μ F bypass capacitor. When placing the capacitor, it should be as close as possible to the power pins VDD1 and VDD2. To eliminate linear reflections, there is a resistor at each end of the cable whose value matches the characteristic impedance of the cable. The bus connectors and termination resistors are as close to the A and B pins as possible.

8.3 ESD Protection

All pins are reinforced with ESD protection construction to prevent encountering electrostatic discharge during handling and assembly. The bus pins have additional anti-static protection for both the logic side (VDD1 side) and the bus side (VDD2 side).

Static protection can be tested in a variety of ways. Here are the ESD specifications for the device.

Bus:

- ± 8 kV HBM.
- ± 12 kV use contact discharge method specified in IEC 61000-4-2

Other pins besides bus pins:

- ± 6 kV HBM.
- ± 7 kV use contact discharge method specified in IEC 61000-4-2

8.4 256 Transceivers Connected to the Bus

The CMT83086 has 1/8 unit load receiver input impedance (96k Ω) and allows up to 256 transceivers to be connected on the bus. Connect any combination of these and other RS-485 devices to the line for up to 32-unit loads.

9 Parameter Test Circuit

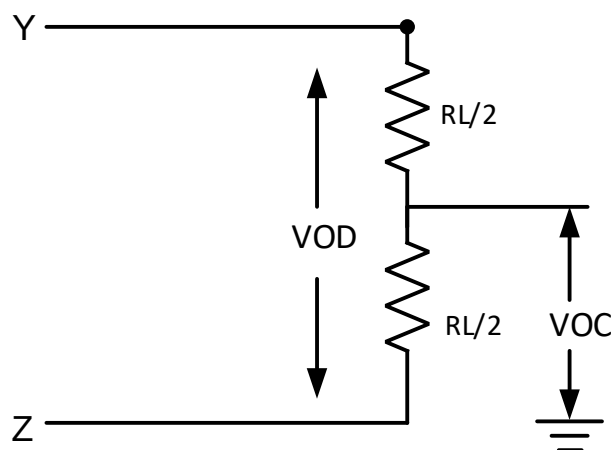


Figure 9-1 Driver V_{OD} Test Circuit

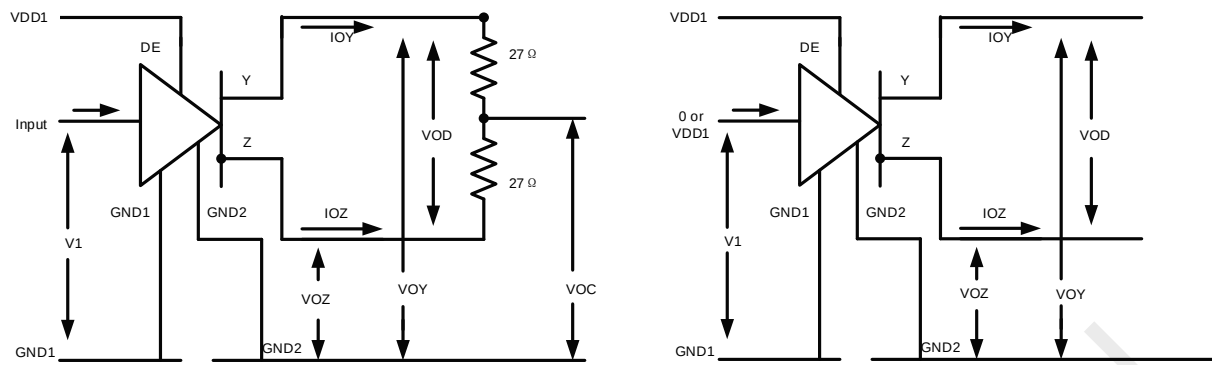
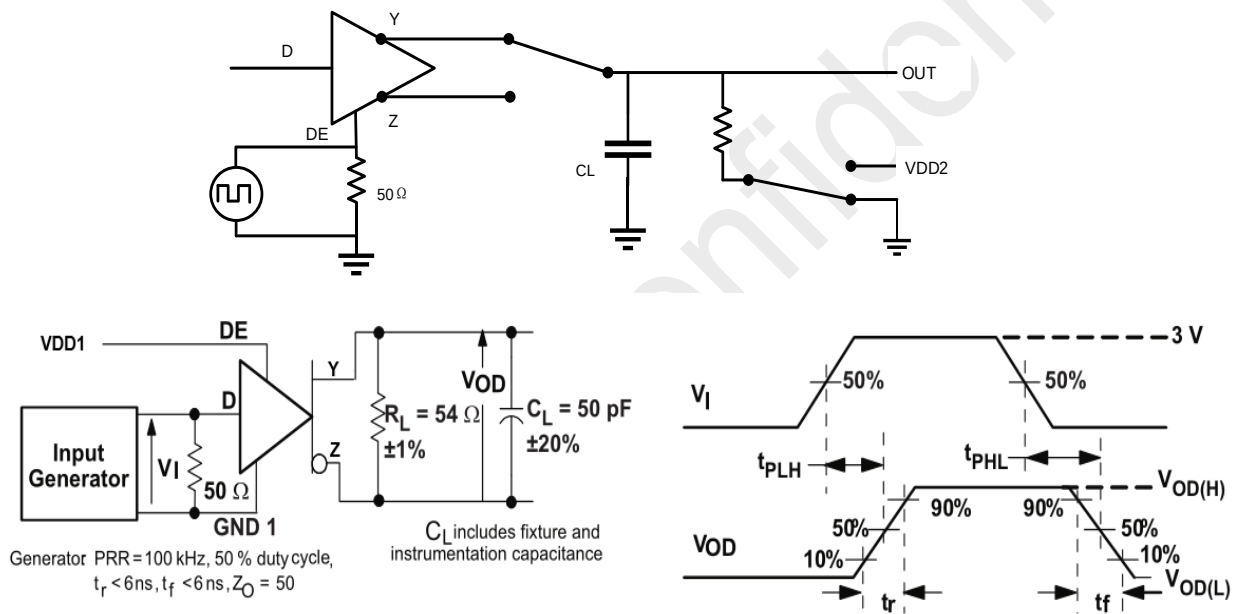
Figure 9-2. Driver V_{OD} Test Circuit and Current Definition

Figure 9-3. Driver Switch Test Circuit and Voltage Waveform

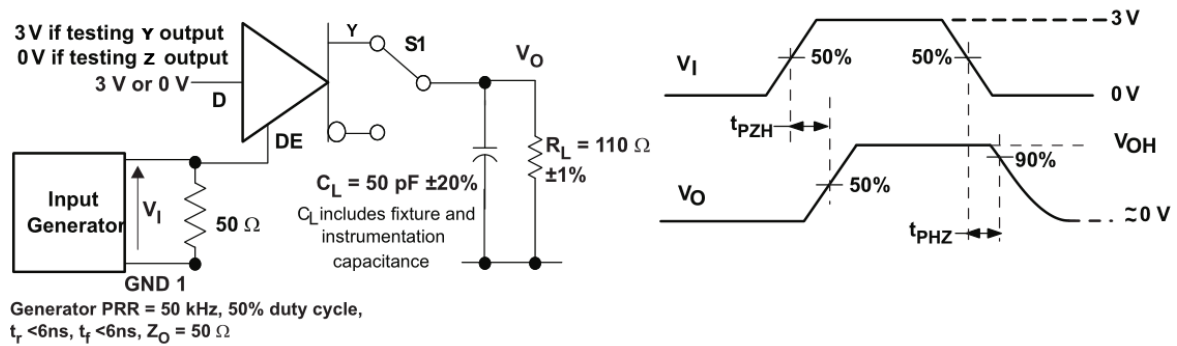


Figure 9-4. Driver Test Circuit and Voltage Waveform of Enable and Disable Time in High Level

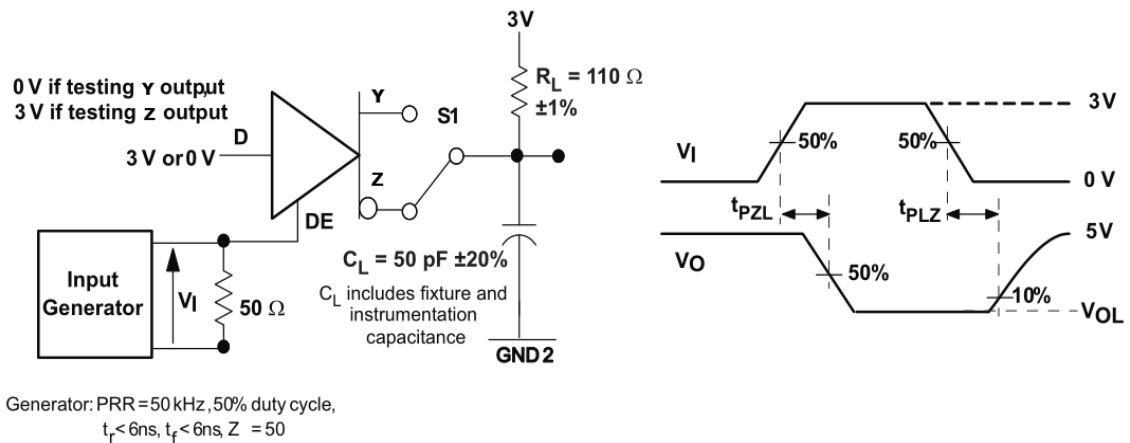


Figure 9-5. Driver Test Circuit and Voltage Waveform of Enable and Disable Time in Low level

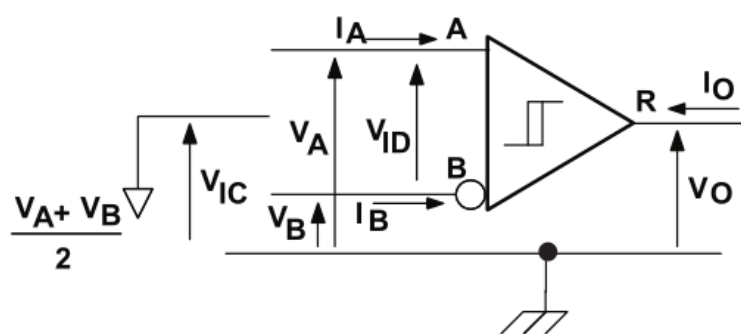


Figure 9-6. Receiver Voltage and Current Definition

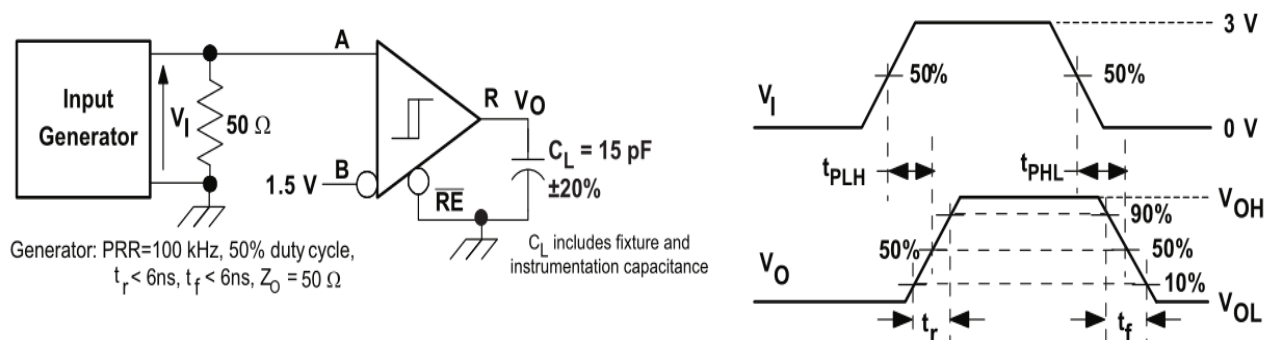


Figure 9-7. Receiver Switch Test Circuit and Waveform

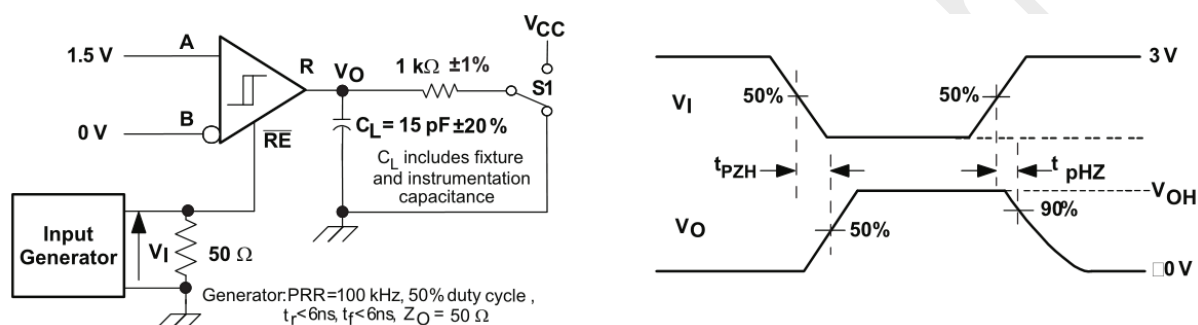


Figure 9-8. Receiver Enables Test Circuit and Waveform, High Level Data Output

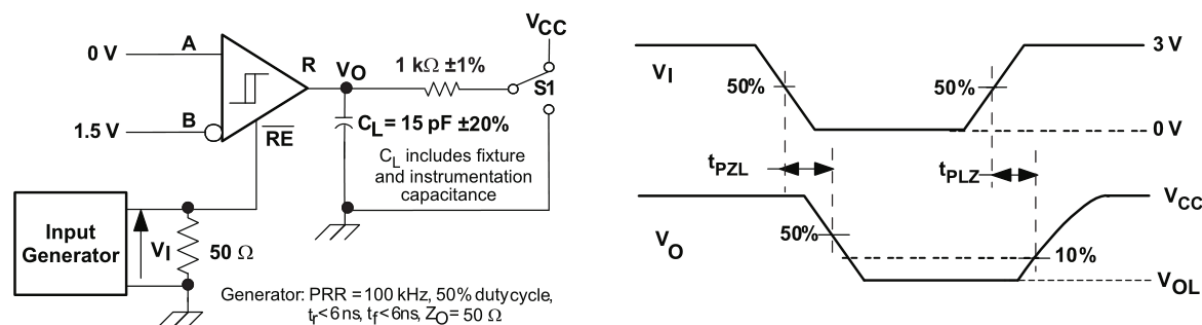


Figure 9-9. Receiver Enables Test Circuit and Waveform, Low Level Data Output

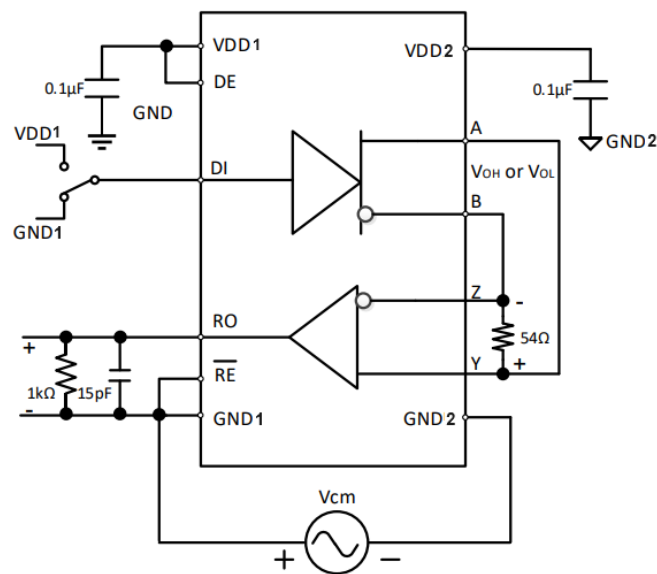
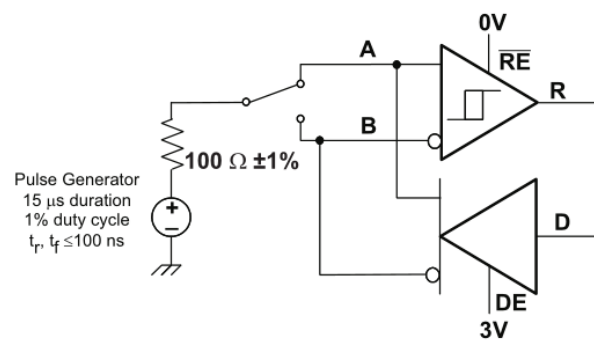


Figure 9-10. Transient Common Mode Rejection Capability Test Circuit



Note: This test is conducted to test survivability only.
Data stability at the R output is not specified.

Figure 9-11 Transient Over-voltage Test Circuit

10 Function Description

10.1 Function Overview

CMT83086 is a high reliability isolated full-duplex RS-485 transceiver. Data isolation is implemented via the Cmostek integrated capacitive isolation, allowing data to be transferred between the logic side and the bus side. CMT83086 is UL1577 safety certified and supports 5kVRMS insulation withstand voltage.

10.2 Data Rate

The maximum data rate of the CMT83086 is 12 Mbps.

10.3 Receiver Input Failure Protection

The device fail safe protection guarantees a receiver logic high output when the receiver input is open or shorted. The receiver

threshold is fixed between -50mV and -200mV, which complies with EIA/TIA-485 standard. If the input differential voltage ($V_A - V_B$) is greater than or equal to - 50 mV, the receiver output R is logic high. If the termination bus and all transmitters are disabled, the differential input voltage will be pulled to zero by the termination resistor. While that receiver output R is logic high based on the receiver threshold.

10.4 Truth Table

Table 10-1. Driver Function Truth Table

V_{DD1}	V_{DD2}	Input (D)	Enable input (DE)	Output (OUTx)	
				Y	Z
PU	PU	H	H	H	L
PU	PU	L	H	L	H
PU	PU	X	L	Z	Z
PU	PU	X	OPEN	Z	Z
PU	PU	OPEN	H	H	L
PD	PU	X	X	Z	Z
PU	PD	X	X	Z	Z
PD	PD	X	X	Z	Z

Table 10-2. Receiver Function Truth Table

V_{DD1}	V_{DD2}	Differential input ($V_A - V_B$)	Enable input (/RE)	Output (R)
PU	PU	≥ -50 mV	L/Open	H
PU	PU	≤ -200 mV	L/Open	L
PU	PU	OPEN/SHORT	L/Open	H
PU	PU	X	H	Z
PU	PU	IDLE	L	H
PD	PU	X	X	Z
PU	PD	X	X	H
PD	PD	X	X	Z

1. PD = Disable; PU = power-on; H = high level; L = low level; X = irrelevant; Z = high impedance.

10.5 Thermal Shutdown

The device protects against overheating damage with an integrated thermal shut down circuit. When the junction temperature (T_J) exceeds + 165 °C (typ), the driver outputs high impedance. When the T_J value is lower than + 145 °C (typ), the equipment resumes normal operation.

11 Package Information

11.1 CMT83086 SOIC-16 Wide Body Package

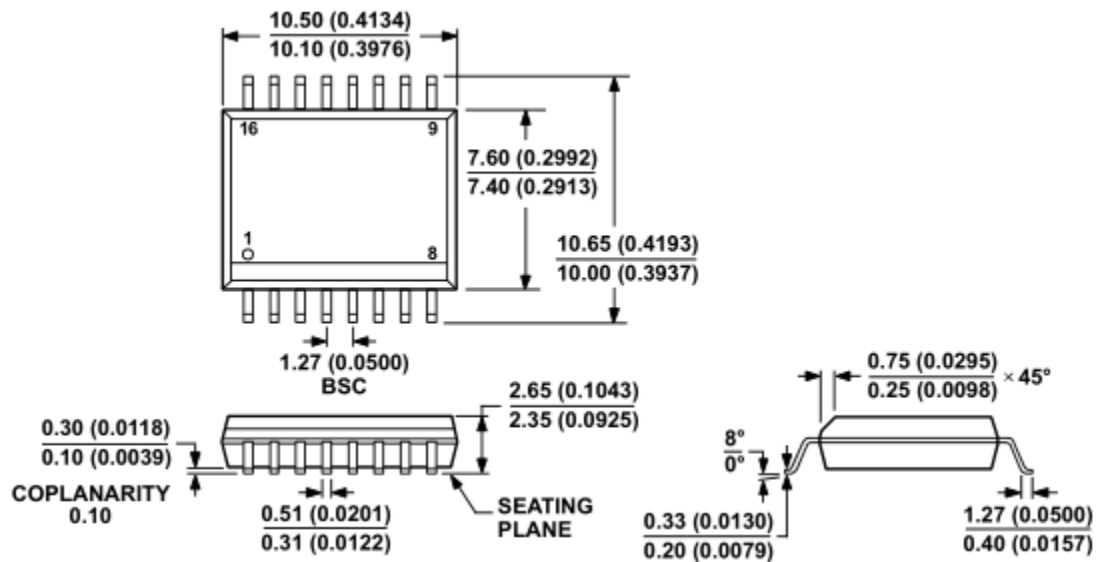


Figure 1. SOIC-16 Wide Body Package

Table 1. SOIC-16 Wide Body Package Dimensions

Symbol	Dimensions (mm)		
	Min	Typ	Max
A	-	-	2.65
A1	0.10	0.20	0.30
A2	2.25	2.30	2.35
A3	1.00	1.05	1.10
b	0.35	0.37	0.43
c	0.15	0.20	0.30
D	10.30	10.40	10.50
E	10.10	10.30	10.50
E1	7.40	7.50	7.60
e	1.14	1.27	1.40
L	0.65	0.70	0.85
L1	1.40		
θ	0	-	8°

12 Ordering information

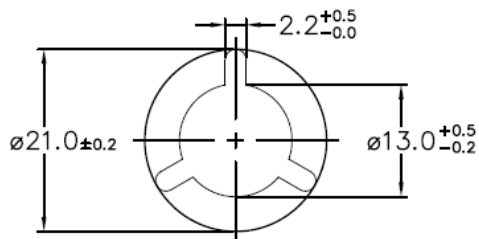
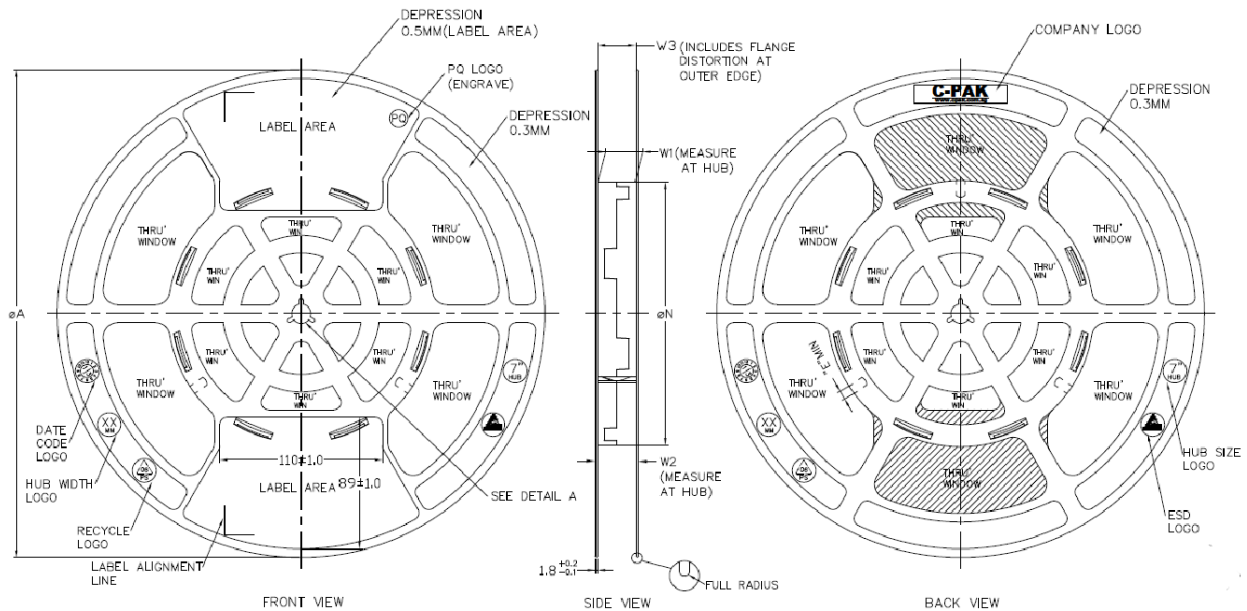
Table 2. Part Number List

Product number	MOQ	Isolation withstand voltage (kV)	Duplex	Node number	MSL	Maximum Data Rate (Mbps)	Temperature range	Package
CMT83086	1000	5	Full duplex	256	2	12	-40 to 125 °C	WB SOIC-16

For more product information, please visit www.hoperf.com.

For ordering requirements, please contact sales@hoperf.com or local sales representative.

13 Tape Information



ARBOR HOLE
DETAIL A
SCALE : 3:1

PRODUCT SPECIFICATION						
TAPE WIDTH	ØA ±2.0	ØN ±2.0	W1	W2 (MAX)	W3	E (MIN)
08MM	330	178	8.4 ^{+1.5} _{-0.0}	14.4	SHALL ACCOMMODATE TAPE WIDTH WITHOUT INTERFERENCE	5.5
12MM	330	178	12.4 ^{+2.0} _{-0.0}	18.4		5.5
16MM	330	178	16.4 ^{+2.0} _{-0.0}	22.4		5.5
24MM	330	178	24.4 ^{+2.0} _{-0.0}	30.4		5.5
32MM	330	178	32.4 ^{+2.0} _{-0.0}	38.4		5.5

SURFACE RESISTIVITY			
LEGEND	SR RANGE	TYPE	COLOUR
A	BELOW 10 ¹²	ANTISTATIC	ALL TYPES
B	10 ⁶ TO 10 ¹¹	STATIC DISSIPATIVE	BLACK ONLY
C	10 ⁵ & BELOW 10 ⁵	CONDUCTIVE (GENERIC)	BLACK ONLY
E	10 ⁸ TO 10 ¹¹	ANTISTATIC (COATED)	ALL TYPES

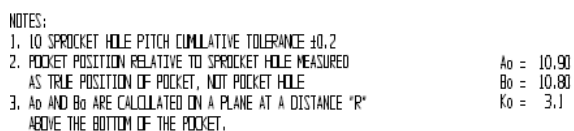


Figure 17. CMT83086 WB SOIC-16 Tape Information

14 Document Revise History

Table 3. Revise Record

Version No.	Chapter	Revise Description	Date
0.1	All	Initial version	2025/3/10

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